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Forecast combination puzzle in the HAR model

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Abstract

The heterogeneous autoregressive (HAR) model has become the benchmark model for predicting realized volatility, given its simplicity and consistent empirical performance. Many modifications and extensions to the original model have been proposed that often only provide incremental forecast improvements. In this paper, we take a step back and view the HAR model as a forecast combination that combines three predictors: previous day realization (or random walk forecast), previous week average, and previous month average. When applying the ordinary least squares (OLS) to combine the predictors, the HAR model uses optimal weights that are known to be problematic in the forecast combination literature. In fact, an average of simpler HAR-style models, and a simple average forecast often outperforms the optimal combination in many empirical applications. We investigate the performance of these simple combination forecasts for the realized volatility of the Dow Jones Industrial Average equity index, and a sample of individual constituent stocks, as well as across a range of other assets, commodities, exchange rates, and a range of global equity market indices. In all cases, we find dramatic improvements in forecast accuracy across all horizons and different time periods. The combinations also perform well relative to the logarithmic form of the HAR model and when using a volatility timing strategy in portfolio allocation. Thisis the first time the forecast combination puzzle is identified in this context.

KEYWORDS

forecast combination, HAR model, realized volatility

1 | INTRODUCTION

Forecasting the volatility of financial asset returns is an important issue in the context of risk management, portfolio construction, and derivative pricing. As such, a great deal of research effort has focused on developing and evaluating volatility forecasting models. With the widespread availability of high-frequency financial data, the recent literature has focused on employing realized volatility (RV) to build forecasting models. The heterogeneous autoregressive (HAR) model of Corsi (2009) was designed to parsimoniously capture the strong persistence typically observed in RV and has become the workhorse of this literature due to its consistently good forecasting performance and that standard linear regression techniques can be used for its estimation.

It is possible to view the HAR model as a combination of three predictors with optimal weighting under ordinary

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least squares (OLS), hence connecting it with the forecast combination literature. The benefits of forecast combinations are well documented; see Timmermann (2006) for an exhaustive literature review and Elliott and Timmermann (2016) or Wang et al. (2023) for a recent and detailed treatment. The M4 competition by Makridakis et al. (2018) shows that the overwhelming majority of the most accurate methods are combinations. It is likely that combinations will become default forecasting methods and benchmarks in the forecasting literature that follows the M4 competition.

All volatility forecasting models, including the HAR, are misspecified (Box, 1979) and hence biased to some degree. Here, we consider the effect of combining a number of simpler forecasts that suffer from less estimation error, combining forecasts based on individual HAR regressors, and a simple average of HAR regressors, whose forecasts are then combined to take advantage of the forecast combination effect. One particularly consistent result is that these combinations of simpler forecasts often outperform more sophisticated alternatives. We find that for both the Dow Jones Industrial Average (DJ) equity index and a sample of individual DJ constituent stocks, these simple combinations dramatically improve forecasting accuracy relative to the traditional HAR model and recent extensions, along with the logarithmic HAR model. These results also translate to other stock market indices and other asset classes. We borrow the "forecast combination puzzle" term to describe this interesting empirical finding from the forecast combination literature. Stock and Watson (2004) were the first to define the puzzle in forecast combinations, but the results were well known before; for example, the equal weights were used in the seminal paper of Bates and Granger (1969) and "wisdom of the crowd" notion goes back centuries (Surowiecki, 2005).

While we provide some possible explanations by drawing on the previous literature, we do not claim to solve the puzzle. The purpose of this paper is to expose this interesting empirical finding and to motivate further research in this direction. For now, the best practical recommendation is to use combinations of these simpler forecasts instead of the OLS estimation in the full HAR model.

The remainder of this paper is organized as follows. In Section 2, we give a brief review of the HAR model together with some recent extensions. Specifically, in Section 2.4, we explain in detail how the HAR model and the forecast combination literature are connected. We provide the empirical evidence of the puzzle in Section 3 and offer some possible explanations. Section 4 provide robustness results in terms of forecast performance relative to the logarithmic HAR model, across different market conditions, and an economic application of the forecasts. Section 5 concludes.

2 | FORECASTING VOLATILITY: HAR MODEL AND ALTERNATIVES

This section provides a brief description of the estimation of RV (and its associated estimation error) upon which the HAR model is based. The forecasting methods considered here are then described. This begins with the well-known HAR model and a recent extension. A description of alternative simpler methods and combination approaches then follows.

2.1 | Measuring volatility

We consider a single asset for which the log-price process P_t within the active part of a trading day evolves in continuous time as

$$dP_t = \mu_t dt + \sigma_t dW_t, \tag{1}$$

where μ_t and σ_t are the instantaneous drift and volatility processes, respectively, and W_t is a standard Brownian motion (Wiener process). The i th Δ-period return within day t is defined as

$$r_{t,i} = P_{t-1+i\Delta} - P_{t-1+(i-1)\Delta},\ i = 1,2,\dots,M,$$

where $M = 1/\Delta$ is the sampling frequency. Hence, the daily return for the active part of trading day t is $r_t = \sum_{i=1}^{M} r_{t,i}$.

In the simplest case, we wish to forecast the latent 1-day integrated variance defined by

$$IV_t = \int_{t-1}^{t} \sigma_s^2 ds. \tag{2}$$

Although (2) is unobservable, it can be consistently estimated by the 1-day realized variance (RV)

$$RV_t = \sum_{i=1}^{M} r_{t,i}^2,$$

as $M \to \infty$ (Andersen & Bollerslev, 1998). Hence, the RV measure is defined as the sum of the squared returns within day t. Given restrictions on the sampling frequency M, Barndorff-Nielsen and Shephard (2002) show that the estimation error in RV can be characterized by $RV_t = IV_t + \eta_t$, where

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$$\frac{\eta_t}{\sqrt{2\Delta IQ_t}} = \frac{RV_t - IV_t}{\sqrt{2\Delta IQ_t}}$$

is approximately $N(0,1)$, or standard normal, and $IQ_t = \int_{t-1}^{t} \sigma_s^4 ds$ is the integrated causticity (IQ), which can be consistently estimated by the realized causticity (RQ)

$$RQ_t = \frac{M}{3} \sum_{i=1}^{M} r_{t,i}^4. \tag{3}$$

2.2 | HAR model

With the widespread availability of high-frequency intraday data, the recent literature has focused on employing RV to build forecasting models for volatility. Among these forecasting models, the HAR model proposed by Corsi (2009) has gained popularity due to its simplicity and consistent forecasting performance in many applications. The formulation of the HAR model is based on a straightforward extension of the so-called heterogeneous ARCH, or HARCH, class of models analyzed by Muller et al. (1997). Under this approach, the conditional variance of the discretely sampled returns is parameterized as a linear function of lagged squared returns over the same horizon together with the squared returns over longer and/or shorter horizons.

The original HAR model specifies RV as a linear function of daily, weekly, and monthly RV components and can be expressed as

$$RV_t = \beta_0 + \beta_1 RV_{t-1}^d + \beta_2 RV_{t-1}^w + \beta_3 RV_{t-1}^m + u_t, \tag{4}$$

where the β_j $(j = 0,1,2,3)$ are unknown parameters that need to be estimated, RV_t is the RV of day t, and $RV_{t-1}^d = RV_{t-1}$, $RV_{t-1}^w = \frac{1}{5}\sum_{i=1}^{5} RV_{t-i}$, $RV_{t-1}^m = \frac{1}{22}\sum_{i=1}^{22} RV_{t-i}$ denote the daily, weekly, and monthly lagged RV, respectively. Corsi (2009) shows that the HAR model provides a simple and tractable approximation to the long-memory observed in RV. Corsi et al. (2012) point out that while it is likely impossible to distinguish between the presence of multiple factors or true long-memory in volatility, it is analytical tractability that is important. Therefore, the HAR model offers this simple analytically tractable approximation to the long-memory in RV, irrespective of the source of the apparent long-memory. In addition to being simply an approximation (of course a very successful one) to an unknown underlying DGP, Buccheri and Corsi (2019) point out that the linear HAR model is misspecified given the nonlinearity in RV.

2.3 | Alternative HAR(like) forecasts

Bollerslev et al. (2016) recently proposed an extension of the HAR model, denoted as the HARQ model, which accounts for the error with which RV is estimated by using RQ. Bollerslev et al. (2016) find that, at least for short-term forecasting, a simplified version

$$RV_t = \beta_0 + \left(\beta_1 + \beta_{1Q}\sqrt{RQ_{t-1}^d}\right) RV_{t-1}^d + \beta_2 RV_{t-1}^w + \beta_3 RV_{t-1}^m + v_t, \tag{5}$$

is useful as most of the attenuation bias in the forecasts (due to RV being less persistent than unobserved IV) is due to the estimation error in RV_{t-1}^d. Overall, this framework allows for less weight to be placed on historical observations of RV when the measurement error captured by RQ is higher.

Given the persistence (strong and slowly decaying autocorrelations) observed in RV series, RV_{t-1}^d, RV_{t-1}^w, and RV_{t-1}^m, the regressors in Equation (4) will likely be strongly correlated leading to increased estimation error. In addition, the HARQ model implies that the importance of the different lagged RV regressors change with estimation error. Therefore, we consider three forecasting regressions (and combinations thereof) only containing individual lagged moving averages of RV,

$$\begin{aligned} RV_t &= \beta_0 + \beta_1 RV_{t-1}^d + v_t^d \\ RV_t &= \beta_0 + \beta_2 RV_{t-1}^w + v_t^w \\ RV_t &= \beta_0 + \beta_3 RV_{t-1}^m + v_t^m. \end{aligned} \tag{6}$$

In addition to forecasts from these three individual models (which we refer to as HARd, HARw, and HARm), these forecasts can be used to construct forecast combinations. Estimation based on the individual models will reduce estimation error partly due to the effect of multicollinearity in the full HAR model. There are other models available in the literature (see Hizmeri et al., 2020), but we will limit ourself to the simplest options to illustrate the forecast combination puzzle in the HAR model.

Alternative forecasts that do not require parameter estimation are random walk forecasts, which are often used as benchmarks for comparing forecast accuracy. Here we will consider three random walk style forecasts (and again, combinations thereof) RV_{t-1}^d, RV_{t-1}^w, and RV_{t-1}^m that are optimally combined in Equation (4). Each of the three predictors are only true random walk forecasts in relation to 1-, 5-, and 22-day ahead forecasts, respectively. While this may be the case, we will consider the performance of the individual predictors and their combinations across different horizons.

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2.4 | Forecast combination view

Here, we consider the HAR model from the forecast combination perspective by considering how to combine it components together in alternative ways. RV_{t-1}^{d}, for example, is a predictor in itself which is provided by the random walk model. RV_{t-1}^{w} is an MA(5) predictor and RV_{t-1}^{m} is an MA(22) predictor. Equation (4) combines the predictors linearly. When OLS is used to estimate the coefficients, it is equivalent to constructing optimal combination; see Granger and Ramanathan (1984). The constant term guarantees the unbiasedness of the combination even when the individual predictions might be biased.

This simple observation allows us to consider other numerous methods to combine the individual predictors that are available in the forecast combination literature. Timmermann (2006) gives an extensive overview of the possible options. One simple method that stands out in particular is simple average. It is a surprisingly robust method that has attracted a lot of attention in empirical studies (see, e.g., Stock & Watson, 2004), and in simulations (see Smith & Wallis, 2009). The fact that the simple average outperforms the optimal combination is often referred to as "forecast combination puzzle."

A theoretical explanation of this puzzle is given in Claeskens et al. (2016), who consider the effect of the estimation of the weights. Estimation of Equation (4) might suffer from various problems: multicollinearity (e.g., during stable times all predictors might be similar), outliers generated by extreme events, and different regimes. These problems that will naturally affect model misspecification and bias, and also estimation error and hence forecast performance. Equally weighted combinations of simpler forecasts such as the univariate models in (6) might be individually biased and not be optimal in theory but significantly reduces the degree of estimation error mitigating the effect of some of these problems.

Here, we consider a number of forecast combinations based on the random walk style predictors (which do not require parameter estimation) and forecasts from the individual regressions in Equation (6). Beginning with the simplest combination of the random walk predictors, a forecast of RV_t is given by the average of the previous daily, weekly, and monthly volatilities, that is,

$$F_t^{\mathrm{sa}} = \frac{1}{3}\left(RV_{t-1}^{d} + RV_{t-1}^{w} + RV_{t-1}^{m}\right). \tag{7}$$

The theoretically optimal (in the MSE sense) forecast of RV_t for the HAR model (4) given the information set at time $t-1$ can be expressed as

$$F_t^{\mathrm{HAR}} = \widehat{\beta}_0 + \widehat{\beta}_1 RV_{t-1}^{d} + \widehat{\beta}_2 RV_{t-1}^{w} + \widehat{\beta}_3 RV_{t-1}^{m}.$$

The simple combination in Equation (7) implies $\widehat{\beta}_0 = 0$ and $\widehat{\beta}_1 = \widehat{\beta}_2 = \widehat{\beta}_3 = 1/3$, restrictions that are unlikely to hold during in-sample estimation but may be beneficial in terms of forecast performance as estimation error is significantly reduced. This approach is denoted as *sa* (simple average).

In addition to the simple average in Equation (7), a combination that accounts for the time-varying estimation error in RV using RQ in a similar vein to the HARQ model is also considered. Weights are computed as

$$w_{t-1}^{d} = \frac{1/\sqrt{RQ_{t-1}^{d}}}{1/\sqrt{RQ_{t-1}^{d}} + 1/\sqrt{RQ_{t-1}^{w}} + 1/\sqrt{RQ_{t-1}^{m}}} \tag{8}$$

with equivalent expressions for weights for the weekly and monthly lags. This rescaling of *RQ* ensures the weights sum to one. The resulting forecast is then given by

$$F_t^{\mathrm{warq}} = w_{t-1}^{d} RV_{t-1}^{d} + w_{t-1}^{w} RV_{t-1}^{w} + w_{t-1}^{m} RV_{t-1}^{m}. \tag{9}$$

This approach will be denoted as *warq* (weighted average using RQ). Equivalent simple average (denoted as *iHARsa*) and RQ-weighted average (*iHARwarq*) combinations of forecasts from the three individual regressions in Equation (6) are also considered. The potential benefit of the latter two approaches, *iHARsa* and *iHARwarq*, is that they weight forecasts from individual models that each suffer from less estimation error than the full HAR model. A range of exogenous variables have been included in HAR models, such as implied volatility or measures of attention and sentiment. One shortcoming of the simpler *sa* combination is that such exogenous variables cannot be included as there is no parameter estimation. There is scope for including exogenous variables into the *iHARsa* regression-based approach. While it is not immediately clear how to achieve this as there is not one full single model, it represents an interesting avenue for future research.

In the forecasting literature, even though the puzzle has been well known for a long time, only recently have we started gaining a better theoretical understanding of it. Claeskens et al. (2016) were the first to show how the weight estimation affects the properties of the combination. If the weights are estimated poorly, the combination might perform worse than the simple average. Gibbs and Vasnev (2018) show the additional information can be

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used to construct conditionally optimal weights that change over time, but the benefits are only visible when the value of this information is greater than the estimation noise. In the case of highly correlated forecasts, Radchenko et al. (2020) show that the balance between the bias and the variance of the weight estimation can be optimized via an additional tuning parameter.

The purpose of this paper is to define the forecast combination puzzle in the HAR model. We do not claim that we solve the puzzle, but we do offer some possible explanations based on the previous literature in the next section.

3 | EMPIRICAL RESULTS

3.1 | Data

The empirical analysis presented here is based on both index level and individual stock data. At the index level, the Dow Jones Industrial Average (DJ) equity index is used, along with 26 individual stocks that are DJ constituents. A list of the individual company names are provided in Table A1 of Appendix A. Estimates of RV_t and RQ_t are based on 5-min intraday DJ returns obtained from Thomson Reuters Datascope. For the DJ data, the sample period spans March 1, 2000, to July 31, 2020 (5058 daily observations). The data for the individual constituent stocks cover the period July 22, 2002, to June 27, 2019 (4193 days). The data that support the findings of this study are openly available in QUT Research Data Finder at https://researchdatafinder.qut.edu.au/individual/n15933.

To give a feel for the data, Figure 1 shows the estimates of DJ RV_t (top panel) and RQ_t (lower panel) for the full sample period. The RV_t series exhibits the well-known features of volatility clustering and is dominated by the periods of high volatility associated with the GFC (2008–2009) and the recent COVID19 pandemic. In comparing the RV_t and RQ_t series, it is clear that they are strongly correlated with the estimation error rising during periods of higher volatility. The main results presented in the body of the paper are based on the DJ index and individual stock data. Additional results will be discussed for other asset classes in the robustness section in Section 4.

3.2 | Forecasting results for DJ and its constituent stocks

The forecasts in the subsequent empirical analysis are generated under a rolling window estimation scheme, where a rolling window of 1000 observations is used for parameter estimation where required. For the DJ case, the out-of-sample period forecast evaluation period begins on April 12, 2004, and for the individual stocks, August 22, 2006. A direct forecasting scheme is used for forecasts for horizons longer than 1 day, an approach commonly applied in the RV forecasting context; see for example, Bollerslev et al. (2016). Under the direct forecasting scheme, average RV over a longer horizon becomes the dependant variable in Equations (4), (6), and (5).

Following the literature on volatility forecast comparison (Patton, 2011; Patton & Sheppard, 2009b), the mean squared error (MSE) and empirical quasi-likelihood (QLIKE) will be used to assess out-of-sample forecast accuracy. For daily RV, they are defined as

$$\mathrm{MSE} = \frac{1}{T}\sum_{t=1}^{T}(RV_t - F_t)^2, \tag{10}$$

and

$$\mathrm{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{RV_t}{F_t} - \log\frac{RV_t}{F_t} - 1\right), \tag{11}$$

where T is the number of forecasts and F_t denotes a forecast of RV_t from the different models or approaches.

Statistically significant differences in forecast performance will be assessed using the model confidence set (MCS) of Hansen et al. (2011). The MCS procedure avoids the specification of a benchmark model and starts with a collection of competing forecasts, $\mathcal{M}^0$, indexed by $i = 1, \ldots, m_0$. Given a loss function, loss differentials $d_{ij,t}$ between forecasts i and j are computed, and H_0: $E(d_{ij,t}) = 0$ for all i,j (the null hypothesis of EPA) is tested. If the null hypothesis is rejected at the significance level α, the worst performing model is eliminated and the process is repeated until nonrejection occurs with the set of surviving models being the MCS, $\hat{\mathcal{M}}^*_{1-\alpha}$. By using the same significance level for all tests, $\hat{\mathcal{M}}^*_{1-\alpha}$ contains the best model(s) from $\mathcal{M}^0$ with a limiting $(1-\alpha)$ level of confidence.[1] In this paper, the results are reported based on the range statistic as described in Hansen et al. (2003).[2] To provide an economic evaluation of the benefit of the combination methods, in addition to the results based on the statistical loss functions (10) and (11), results for a portfolio allocation application are also reported in the robustness section in Section 4.

Table 1 reports the ratios in MSE loss for each of the forecasting approaches relative the HAR model across 1-, 5-, 10-, and 22-day forecast horizons for the DJ dataset. *

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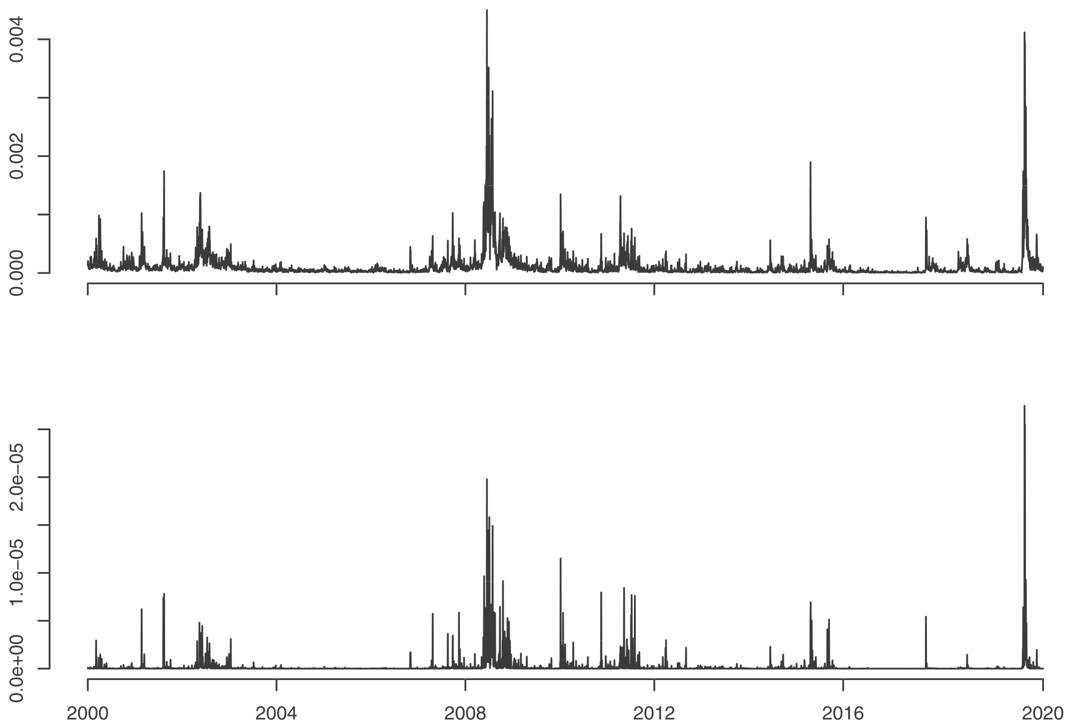

FIGURE 1 Estimates of RV_t (top panel) and RQ_t (lower panel) for the DJ over the sample period March 1, 2000, to July 31, 2020 (5058 daily observations).

indicates that a forecast is included in the MCS at a 10% level of significance. While not the central focus here, the HARQ model offers reductions in loss (relative to the HAR) at the shorter horizons. The results for the individual models in the next three rows show that there are some benefits from HARd, that is, using only RV^d_{t-1}, at the slightly longer horizons. However, in all other cases, there are no clear benefits. While there may be some benefits of using shorter term lags in simpler models to reduce estimation error, this does come at the cost of forecast accuracy in many instances. The next block shows the results for the three random walk style models that involve no estimation. An interesting result is observed from using RV^w_{t-1} as a forecast as it offers gains in accuracy even though no estimation is involved. In contrast, the daily and monthly MAs in RV do not offer consistent benefits. These results indicate that even without estimation, some degree of smoothing is beneficial for forecasting across different horizons.

The final block of results considers combinations of the individual models in Equation (6) and the random walk models. Starting with the equally weighted combination of the individual regressions (iHARsa), consistent gains in loss across the forecast horizons are observed. While there are some small gains from the individual regressions, a reduction in estimation error combined with the benefit of forecast combinations leads to consistent gains. The unequally weighted combination of the individual models driven by estimation error (iHARwarq) offers incremental improvements over the equally weighted case. This shows that accounting for the estimation error in the underlying volatility is as small degree of benefit. Finally, moving to the combinations of the random walks (sa and warq) reveals the largest gains in forecast accuracy. While the individual random walk forecasts offered some gains, a simple average (sa) outperforms any of the regression based forecasts. Again, combining on the basis of the estimation error in RV, using RQ, (warq) offers small gains but much less than in the previous case. While the combination methods do provide large gains in average MSE loss, it is not possible to statistically distinguish between the forecasts under MSE using the MCS. All forecasts are members of the MCS at a 10% level of significance. This result is consistent with the findings of Patton and Sheppard (2009a) who note the inability of the MSE loss function to select between forecasts relative to the QLIKE loss function.

Table 2 reports the equivalent DJ results under the QLIKE loss function. While QLIKE is clearly a very different loss function than MSE, there are important similarities with the MSE results. There are no consistent gains from the individual regressions or random walk forecasts (RV^w_{t-1} is the best performer out of these). While the *iHARsa* combination does not perform as well under

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TABLE 1 DJI forecasting performance for different approaches using the rolling window MSE ratios relative to the HAR model for the whole 2000–2020 period. ∗ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000∗	1.000∗	1.000∗	1.000∗
HARQ	0.943∗	0.977∗	1.073∗	1.126∗
HARd	1.136∗	0.788∗	0.730∗	0.835∗
HARw	1.001∗	1.031∗	1.015∗	0.851∗
HARm	1.463∗	1.664∗	1.631∗	1.792∗
Forecasts based on lags				
RV^d_{t-1}	1.165∗	0.947∗	0.840∗	1.282∗
RV^w_{t-1}	0.945∗	0.727∗	0.617∗	0.959∗
RV^m_{t-1}	1.410∗	1.056∗	0.793∗	1.037∗
Forecast combinations				
iHARsa	0.972∗	0.865∗	0.846∗	0.923∗
iHARwarq	0.952∗	0.837∗	0.791∗	0.878∗
sa	0.902∗	0.638∗	0.546∗	0.847∗
warq	0.897∗	0.629∗	0.529∗	0.786∗

TABLE 2 DJI forecasting performance for different approaches using the rolling window QLIKE ratios relative to the HAR model for the whole 2000–2020 period. ∗ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000	1.000	1.000∗	1.000∗
HARQ	3.017	1.014	1.836	0.973∗
HARd	1.260	1.344	1.278	1.133
HARw	1.100	1.112	1.064	1.017∗
HARm	1.384	1.377	1.278	1.124
Forecasts based on lags				
RV^d_{t-1}	1.340	1.613	1.781	1.819
RV^w_{t-1}	1.049	1.054	1.083	1.196
RV^m_{t-1}	1.350	1.329	1.247	1.179
Forecast combinations				
iHARsa	1.142	1.161	1.111	1.036∗
iHARwarq	1.134	1.193	1.154	1.074
sa	0.962∗	0.919∗	0.957∗	1.051∗
warq	1.044	1.149	1.265	1.366

QLIKE, the simple average of the random walks (*sa*) is again the best performer overall. This result is confirmed as the sa forecast is the only member of the MCS at all forecast horizons, and is the sole member at the 1- and 5-day horizons. Tables B1 and B2 in Appendix B report equivalent results for a rolling estimation window of 500 observations. The results are very similar to those discussed above, indicating the benefit of the combination methods is robust to shorter estimation windows.

Equivalent full sample forecasting results for the individual stocks are reported in Tables B6 and B7 provided in Appendix B. These are reported in the form of average ratios (relative to HAR) across the 26 stocks, with the number times each forecast is included in the MCS reported in the superscripts. Overall, these paint a very similar picture to those based on the DJ index. The simple average of the random walks (*sa*) continues to provide reductions in loss under both MSE and QLIKE, with the *iHARsa* combination continuing to work well under MSE. Overall, the MCS results also paint a similar picture. While large reductions in MSE loss are observed, most forecasts are contained in the MCS across most of the 26 stocks. Under QLIKE, the simple *sa* combination is a member of the MCS in the majority of cases.

The empirical results show that simple combination forecasts dramatically outperform the traditional HAR model and its modifications. Following the forecast combination literature, we identify this empirical result as the forecast combination puzzle in the HAR model. This is the first time the puzzle is identified in realized volatility forecasts.

3.3 | The role of estimation error

All of the forecasting models are biased to some degree, including the full HAR model, which is only an approximation (of course, successful one) to an unknown process driving the apparent long-memory in RV. Moving beyond the bias, reduced estimation error may partly explain the success of the combinations of the simpler forecasts.

Table 3 reports the standard errors for the HAR model in Equation (4), and the individual models in Equation (6) for the full sample of data across 1-, 5-, 10-, and 22-day forecast horizons. The ratio of the standard error from the individual model relative to the corresponding lag in the full model is shown in the last column. This gives an indication of the effect on estimation error by avoiding estimating all HAR coefficients. The reported ratios in the final column show that 30% to 60% reductions in estimation error are observed when estimating the models individually. This reduction in estimation error may explain some of the improvements observed with the *iHARsa* forecast. By extrapolating this logic to the random walk forecasts that do not require coefficient estimation (and therefore omitted in Table 3), we may explain their success. The fact the simple average of the random walks (*sa*) performs particularly well during periods of high volatility (see Table B3 and B5 in Appendix B), when estimation error (including in RV itself) is high, lends support to the importance of reducing estimation error.

TABLE 3 Standard errors of estimated coefficients over the whole period from the full HAR model in Equation (4) and models based on individual predictors (HARd, HARw, and HARm) in Equation (6). The ratio of the standard error from the individual model relative to the corresponding lag in the full model is shown in the last column.

	HAR	HARd	HARw	HARm	Ratio
Forecasting horizon: 1-day					
β_1	0.0171	0.0094			0.5499
β_2	0.0255		0.0102		0.3980
β_3	0.0204			0.0140	0.6869
Forecasting horizon: 5-day					
β_1	0.0140	0.0078			0.5578
β_2	0.0209		0.0084		0.4033
β_3	0.0167			0.0118	0.7027
Forecasting horizon: 10-day					
β_1	0.0145	0.0080			0.5530
β_2	0.0217		0.0086		0.3964
β_3	0.0174			0.0115	0.6598
Forecasting horizon: 22-day					
β_1	0.0154	0.0083			0.5384
β_2	0.0235		0.0091		0.3843
β_3	0.0199			0.0115	0.5785

Figure 2 shows the estimated HAR coefficients for the full HAR model through time (dotted lines) and coefficients from the individual regressions in Equation (6) (dashed lines) scaled by 1/3 as the *iHARsa* combination forecast is an average of these three models. As a point of reference, these are shown against a fixed level of 1/3 (solid horizontal lines). It is clear that the estimated coefficients for the full HAR model undergo dramatic and rapid change, indicating the weighting of the different lags in RV is changing. While some parameter variation is to be expected, certainly if there are structural changes in process underlying RV, the degree of variation in the rolling coefficients is stark. Interestingly, $\hat{\beta}_1$ and $\hat{\beta}_2$ fluctuate around means close to 1/3 (weights under the equal weighted simple average, *sa*). The coefficients from the three individual regressions in Equation (6) exhibit much less variation than those from the full HAR model, reflecting less estimation error, while remaining closer to 1/3. The empirical results in Section 3.2 show that this variation of full HAR coefficients around the fixed 1/3 alternative, or those from the *iHARsa* model, offers no benefits in terms of out-of-sample forecasting. Seemingly, increased estimation error under the full HAR model outweighs any benefit from changes in coefficients reflecting changes in dynamics.

4 | THE EXTENT OF THE PUZZLE: ROBUSTNESS CHECKS

This section reports the results for a number of robustness checks across different issues, different benchmark forecasts, different market conditions, application to different assets, and performance under portfolio allocation.

4.1 | The logarithmic HAR benchmark

Early work on modeling and forecasting RV worked with $\ln(RV_t)$ (Andersen et al., 2001, 2003), an approach often used as a benchmark forecast. Therefore, as a robustness check, this section presents results comparing the simple combinations of lagged RV to the logarithmic HAR. The logarithmic form of the HAR model is given by

$$\ln(RV_t) = \beta_0 + \beta_1 \ln\left(RV_{t-1}^d\right) + \beta_2 \ln\left(RV_{t-1}^w\right) + \beta_3 \ln\left(RV_{t-1}^m\right) + u_t. \tag{12}$$

The explanatory variables here are constructed in the same manner as discussed earlier, with moving averages

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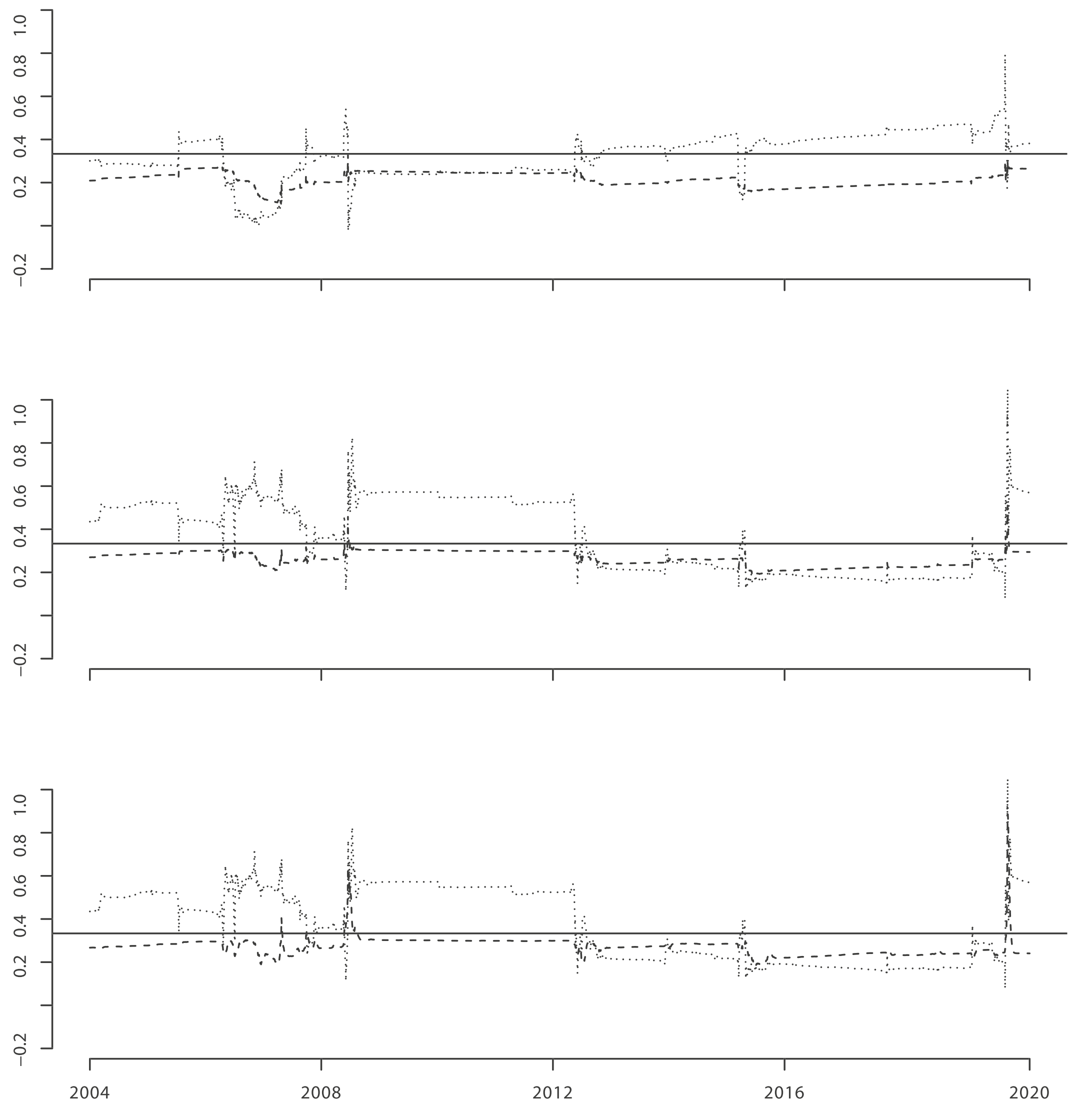

FIGURE 2 Dynamics of estimated coefficients. Full HAR model coefficients in Equation (4) (dotted lines) and three individual regressions in Equation (6) (dashed lines). Top panel: $\hat{\beta}_1$, middle panel: $\hat{\beta}_2$, bottom panel: $\hat{\beta}_3$. All coefficients are estimated using a 1000-day rolling window and 1-day forecasting horizon. The solid horizontal lines represent a value of 1/3.

of $\ln(RV_t)$ taken. From tab. 1 in Proietti and Lütkepohl (2013), a bias-corrected forecast of RV_t for the HAR model applied to *logarithmic* (instead of raw) daily RV is

$$F_t = \exp\left\{\beta_0 + \beta_1 \log RV_{t-1} + \frac{\beta_2}{5}\sum_{i=1}^{5} \log RV_{t-i} + \frac{\beta_3}{22}\sum_{i=1}^{22} \log RV_{t-i} + \frac{\sigma_t^2}{2}\right\}, \tag{13}$$

where σ_t^2 is the conditional variance of the errors u_t.

Here, logarithmic versions of *HARd*, *HARw*, and *HARm* from Equation (6) will be considered, denoted as *log-HARd*, *log-HARw*, and *log-HARm*. The simple equally weighted combination of these, *log-iHARsa* is considered. To highlight the potential benefit of the simple combination, the random walk forecasts, RV_{t-1}^d, RV_{t-1}^w, and RV_{t-1}^m along with their simple combination *sa*, continue to be based on raw RV.

Table 4 reports the ratios in MSE loss for each of the forecasting approaches relative the log-HAR model across 1-, 5-, 10-, and 22-day forecast horizons for the DJ dataset, again with $*$ indicating a forecast is included in the MCS at a 10% level of significance. The top panel of results show that under the logarithmic model, there are no consistent gains in MSE loss from the models containing the individual regressors, *log-HARd*, *log-HARw*, and *log-HARm*. This is a similar pattern to the previous results. The next block of results show that the three random walk style models based on raw RV (involving no estimation) offer no gains relative to the logarithmic

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TABLE 4 DJI forecasting performance for different approaches using the rolling window MSE ratios relative to the log-HAR model for the whole 2000–2020 period. $*$ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: log-HAR in Equation (12), logarithmic HAR with only daily (log-HARd), weekly (log-HARw) or monthly (log-HARm) predictors in Equation (6). Forecast combinations: log-iHARsa—average of logarithmic HAR models with individual predictors, sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from log-HAR and related models				
log-HAR	1.000*	1.000*	1.000*	1.000*
log-HARd	1.231*	1.298*	1.257*	1.174*
log-HARw	1.078*	1.072*	1.025*	0.996*
log-HARm	1.762*	1.643*	1.389*	1.207*
Forecasts based on lags				
RV^d_{t-1}	1.291*	1.318*	1.398*	1.597*
RV^w_{t-1}	1.048*	1.012*	1.027*	1.195*
RV^m_{t-1}	1.563*	1.471*	1.320*	1.292*
Forecast combinations				
log-iHARsa	1.167*	1.175*	1.104*	1.048*
sa	1.000*	0.889*	0.908*	1.054*

TABLE 5 DJI forecasting performance for different approaches using the rolling window QLIKE ratios relative to the log-HAR model for the whole 2000–2020 period. $*$ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: log-HAR in Equation (12), logarithmic HAR with only daily (log-HARd), weekly (log-HARw) or monthly (log-HARm) predictors in Equation (6). Forecast combinations: log-iHARsa—average of logarithmic HAR models with individual predictors, sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from log-HAR and related models				
log-HAR	1.000*	1.000*	1.000*	1.000*
log-HARd	1.153	1.072	1.017	0.995*
log-HARw	1.096	1.025	0.964	0.967*
log-HARm	2.395	1.864	1.464	1.167
Forecasts based on lags				
RV^d_{t-1}	1.230	1.110	1.203	1.254
RV^w_{t-1}	1.159	0.979	0.925*	0.999*
RV^m_{t-1}	2.077	1.604	1.263	1.092
Forecast combinations				
log-iHARsa	1.140	1.024*	0.963*	0.949*
sa	1.036	0.882*	0.867*	0.927*

model. Relative to the earlier results, this difference in relative performance is unsurprising as the log-HAR model is a more competitive benchmark in terms of forecast performance. Results in the bottom panel show that combination of the individual log-HAR models, *log-iHARsa* offers no clear benefit. On the other hand, the simple average of the RV lags, sa provide forecasts that are at least as good as the log-HAR for the 1-, 5-, and 10-day horizons. This a very interesting result in that the benefit of simple averaging with no estimation often dominates even the more competitive logarithmic HAR model. Again here, it is not possible to distinguish between the forecasts using the MCS under MSE loss.

Table 5 provides the equivalent results under the QLIKE loss function. Overall, these paint a very similar picture to those reported in Table 4. Under QLIKE, the simple average sa forecast performs at least as well as log-HAR. The combination of individual log-HAR models, log-iHARsa also performs well under QLIKE. The MCS results show that the simple combinations, including the simplest sa forecast are included in the MCS at all forecast horizons longer than 1-day ahead. This shows that even the simple sa forecast is of equal predictive accuracy relative to the competitive logarithmic HAR.

Tables B8 and B9 (in Appendix B) report the equivalent results relative to the log-HAR for the sample of individual stocks. The findings from the individual stocks are consistent with those reported here the DJ index.

4.2 | Changing market conditions

How the relative performance of the combination forecasts changes over time with changing market conditions will be considered in two ways. First, Figure 3 shows the cumulate MSE (top panel) and QLIKE loss (bottom panel) for the 1-day ahead forecasts. Here, the full HAR model (solid line) is compared with the *iHARsa* (dashed line) and *sa* (dotted line) forecasts to gain a deeper understanding of the benefits of the combination methods. The top panel shows that both combination methods experience smaller increases in cumulative loss relative to the full HAR during periods of crisis, GFC in 2008–2009 and COVID in 2020. This shows that the combination methods work well during periods of market turmoil, which correspond to periods when the coefficients in the full HAR model are changing significantly, see Figure 2. The bottom panel shows that QLIKE loss is not sensitive to periods crisis. The sa and iHARsa forecasts slightly outperform and underperform the HAR, respectively, consistently through time.

Additional robustness results (Tables B3,B4, and B5 provided in Appendix B) examine how MSE loss changes across subperiods of differing levels of volatility, 2008–2011 (high volatility period containing the GFC), 2012–2018 (low volatility), and a short period 2019–2020 (high volatility containing the COVID-19 pandemic). These results show that much of the benefit of the combination methods is driven by periods of high volatility. During the 2012–2018 period, the combination methods provided similar performance to the full HAR model. These results are consistent with the behavior of the cumulative loss shown in Figure 3.

4.3 | Forecasting results across other asset classes

The results presented so far have been based on one equity index and individual stocks. Here, we consider the benefits of the combination methods across a range of other assets, commodities (USO, United States Oil ETF, GLD, SPDR Gold Trust ETF and SLV, iShares Silver Trust ETF), exchange rates (EUR/USD and GBP/USD), and a range of global equity market indices. More details regarding the samples and selected results are presented in Appendix B; see Tables B10–B16.

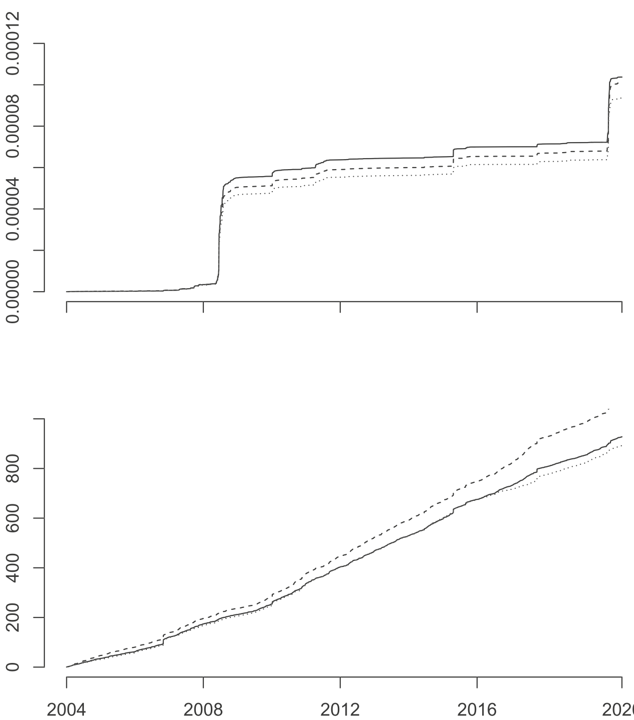

FIGURE 3 Cumulative loss for HAR (solid line), iHARsa (dashed line) and sa (dotted line). Top panel: MSE loss. Bottom panel: QLIKE loss.

In the case of crude oil, the *iHARsa* and *sa* combination forecasts offer similar gains in performance in comparison to the results presented earlier. For the other commodities, GLD and SLV, the combinations provide no large gains relative to the full HAR model, but the performance is equivalent to the HAR in most cases. Results for EUR/USD exchange rate are similar to the GLD and SLV cases. In the case of GBP/USD, the combination methods do offer significant gains over the full HAR, but not to the same magnitude as in the case of equities. Results for other equity market indices (not reported here, but available upon request), Australian All Ordinaries, FTSE 100, DAX, Hang Seng, Nikkei 225, and EURO-STOXX 50, all follow the same pattern as those presented earlier for the DJ index and the individual stocks. Overall, there are benefits from the two simple combination methods across a wide range of assets. While they do not outperform the full HAR in every case, they rarely offer inferior performance. While significant gains in performance are observed in a number of the assets (USO crude oil and GBP/USD), the largest gains are observed in equity markets. Relative to most other asset classes, equities undergo the largest, and often most rapid changes in the level of volatility. It has been shown earlier that the combinations perform well during these periods of instability and hence provide an explanation of why the combinations perform best in the equity context. That being said, the combination forecasts are rarely inferior across a wider range of settings beyond equities, which in itself is remarkable.

4.4 | Portfolio allocation

Given the univariate volatility forecasts considered here, the volatility timing (VT) framework of Kirby and Ostdiek (2012) is used to examine the economic benefit of the proposed combination methods. As opposed to traditional mean-variance optimization, Kirby and Ostdiek (2012) propose a simple volatility timing strategy which involves no optimization nor matrix inversion. This approach was originally designed to reduce estimation error and portfolio turnover which reduces transactions costs. It assumes that the off-diagonal elements of the covariance matrix are zero, which can be interpreted as an aggressive form of shrinkage, meaning only univariate volatility forecasts are required. This VT application is based on the same sample of 26 individual stocks discussed in Section 3.1. Given a forecast of RV for asset i, $\widehat{F_t^i}$, the weight placed in asset i in period t under VT, is defined as

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TABLE 6 Volatility of portfolio returns in annualized terms.

	HAR	iHARsa	sa
	Volatility of $r_{p,t}$		
$\eta=1$	12.50%	12.52%	12.51%
$\eta=2$	11.64%	11.63%	11.63%
$\eta=4$	11.08%	10.96%	11.03%
	Turnover: $\overline{TO}$		
$\eta=1$	0.1108	0.0676	0.1304
$\eta=2$	0.2072	0.1264	0.2451
$\eta=4$	0.3641	0.2219	0.4252

$$\mathrm{w}_{i,t}=\frac{\left(1/\widehat{F}_t^i\right)^\eta}{\sum_{i=1}^{N}\left(1/\widehat{F}_t^i\right)^\eta}, \tag{14}$$

where η is a tuning parameter that controls the responsiveness of the weights to changes in the variances. Larger weight is placed into assets with forecasts of lower variance, the η controlling the aggressiveness with which the weights respond to differences in variances.

Given the portfolio weights, returns are then defined as

$$r_{p,t}=\mathbf{w}_t\mathbf{r}_t' \tag{15}$$

where $\mathbf{w}_t$ and $\mathbf{r}_t$ are vectors containing the weights and returns for the individual stocks.

Changing forecasts results in changing weights or portfolio turnover. Reduced turnover is preferred to minimize transaction costs. Volatility in portfolio returns and formal portfolio turnover are used to consider the economic benefit of the combination-based forecasts. A formal portfolio turnover measure is defined as follows. To begin, if \$1 is invested in the portfolio at time $t-1$, asset i will have a value of $\mathrm{w}_{i,t-1}(1+r_{i,t})$ at time t. Therefore, the weight in asset i before rebalancing at time t is

$$\tilde{\mathrm{w}}_{i,t}=\frac{\mathrm{w}_{i,t-1}(1+r_{i,t})}{\sum_{i=1}^{N}\mathrm{w}_{i,t-1}(1+r_{i,t})}, \tag{16}$$

with the turnover at time t given by

$$TO_t=\sum_{i=1}^{N}|\mathrm{w}_{i,t}-\tilde{\mathrm{w}}_{i,t}|+\left|\sum_{i=1}^{N}(\mathrm{w}_{i,t}-\tilde{\mathrm{w}}_{i,t})\right|. \tag{17}$$

To highlight the economic benefit of the combination forecasts, the performance of the HAR, *iHARsa*, and *sa* 1-day forecasts are compared. Forecasts of RV for each of the 26 individual stocks are used to construct portfolio weights from Equation (14), returns from Equation (15) and finally turnover from Equations (16) and (17).

Table 6 contains the results for the volatility of portfolio returns and portfolio turnover for values of $\eta=1,2$, and 4. It is clear the choice of forecasting model has virtually no impact on the volatility of portfolio returns. While the combination methods *iHARsa* and *sa* produce less volatile forecasts of RV (results not shown here, but available upon request) relative to HAR, this has no impact on the volatility of the final portfolio returns. When average turnover is considered, the benefit of the *iHARsa* combination method becomes apparent. Relative to the full HAR, the combination *iHARsa* forecasts lead to reductions in portfolio turnover of nearly 40%, while achieving the same volatility in portfolio returns. This is consistent with reduction in coefficient variability observed in Figure 2. The simple *sa* combination leads to small increases in turnover. While the *sa* forecasts (along with the *iHARsa* forecasts) are less volatile than those from HAR, the *sa* forecasts are on average slightly lower than HAR and *iHARsa*. Given the portfolio weights $\mathrm{w}_{i,t}$ are based on the inverse of the forecasts, small reductions in the mean of the RV forecasts lead to increases in the level and variability in the portfolio weights as differences between the assets are magnified. Overall, these results show that trading costs could be reduced by using a combination forecast from the three simpler regression based forecasts in the form of the *iHARsa* approach as opposed to a full HAR model.

5 | CONCLUSION

Despite recent advances, there is still a lot to understand about the forecast combination puzzle in the traditional forecasting literature. In fact, in more than 50 years of forecast combination literature with the diverse set of forecast combination weights, "forecasters still have little guidance on how to solve the forecast combination puzzle" as noted in a recent review by Wang et al. (2023). In our paper, we introduce the forecast combination puzzle within the context of the HAR model. It is shown that two simple combination-based alternatives to the HAR often outperform. Combining forecasts from simpler HAR style regressions or across the HAR regressors themselves often outperform the HAR model itself. This improvement is due to minimizing, or avoiding, estimation error when dealing with correlated regressors and parameter instability when forecasting through periods of turmoil. These approaches perform well under statistical loss, across a range of assets. They can also offer improvements in portfolio stability in an economic loss

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setting. While we provide some explanations drawing from the previous literature, we do not claim to have completely solved the puzzle. Our goal is to expose this interesting empirical finding and to motivate further research. However, once a deeper understanding of the puzzle emerges, more sophisticated solutions will follow.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in QUT Research Data Finder at http://researchdatafinder.qut.edu.au/individual/n15933.

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ENDNOTES

[1] In this sense, the MCS at level α is similar to a $(1-\alpha)\%$ confidence interval for an unknown parameter.

[2] MCS code from the MFE toolbox of Kevin Sheppard, https://www.kevinsheppard.com/MFE_Toolbox, was used.

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APPENDIX A: LIST OF INDIVIDUAL STOCKS

TABLE A1 Ticker symbols and company names for the 26 NYSE stocks.

Symbol	Company
AAPL	Apple Inc.
AXP	American Express
BA	Boeing Co.
CAT	Catepillar Inc.
CSCO	Cisco Systems Inc.
CVX	Chevron Corp.
DIS	Walt Disney Co.
GS	Goldman Sachs Group Inc.
HD	Home Depot Inc.
IBM	IBM
INTC	Intel Corp.
JNJ	Johnson & Johnson
JPM	JP Morgan Chase & Co.
KO	Coca-Cola Co.
MCD	Mcdonald's Corp.
MMM	3M Co.
MRK	Merck & Co. Inc.
MSFT	Microsoft Corporation
NKE	Nike Inc.
PFE	Pfizer Inc.
PG	Procter & Gamble Co
UNH	UnitedHealth Group Inc.
UTX	United Technologies Corp.
VZ	Verizon Communications Inc.
WMT	Walmart Inc.
XOM	Exxon Mobil Corporation

APPENDIX B: ADDITIONAL EMPIRICAL RESULTS

TABLE B1 DJI forecasting performance for different approaches using the rolling window (500 observations) MSE ratios relative to the HAR model for the whole 2000–2020 period. $*$ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000$*$	1.000$*$	1.000$*$	1.000$*$
HARQ	0.922$*$	0.966$*$	0.953$*$	1.159$*$
HARd	1.111$*$	0.680$*$	0.660$*$	0.845$*$
HARw	0.988$*$	0.973$*$	1.054$*$	0.867$*$
HARm	1.440$*$	1.587$*$	1.682$*$	2.007$*$
Forecasts based on lags				
RV^d_{t-1}	1.127$*$	0.806$*$	0.693$*$	1.278$*$
RV^w_{t-1}	0.919$*$	0.620$*$	0.509$*$	0.956$*$
RV^m_{t-1}	1.367$*$	0.897$*$	0.652$*$	1.030$*$
Forecast combinations				
iHARsa	0.959$*$	0.813$*$	0.847$*$	0.984$*$
iHARwarq	0.935$*$	0.779$*$	0.769$*$	0.922$*$
sa	0.875$*$	0.544$*$	0.450$*$	0.842$*$
warq	0.870$*$	0.536$*$	0.436$*$	0.784$*$

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TABLE B2 DJI forecasting performance for different approaches using the rolling window (500 observations) QLIKE ratios relative to the HAR model for the whole 2000–2020 period. ∗ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000	1.000	1.000	1.000∗
HARQ	1.319	1.707	0.953∗	1.084
HARd	1.204	1.296	1.170	1.110
HARw	1.070	1.110	1.002	1.011∗
HARm	1.355	1.397	1.228	1.155
Forecasts based on lags				
RV^d_{t-1}	1.322	1.582	1.612	1.755
RV^w_{t-1}	1.022	1.023	0.974	1.147
RV^m_{t-1}	1.309	1.289	1.122	1.136
Forecast combinations				
iHARsa	1.107	1.153	1.046	1.041
iHARwarq	1.107	1.194	1.093	1.082
sa	0.942∗	0.895∗	0.862∗	1.010∗
warq	1.022	1.116	1.136	1.310

TABLE B3 DJI forecasting performance for different approaches using the rolling window MSE ratios relative to the HAR model for the 2008–2011 period (high volatility).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000	1.000	1.000	1.000
HARQ	0.858	0.956	0.988	1.158
HARd	1.195	1.023	0.852	0.719
HARw	0.963	0.909	0.746	0.737
HARm	1.206	1.366	1.402	1.678
Forecasts based on lags				
RV^d_{t-1}	1.203	1.264	1.104	0.966
RV^w_{t-1}	0.917	0.810	0.660	0.599
RV^m_{t-1}	1.156	0.988	0.740	0.666

(Continues)

TABLE B3 (Continued)

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecast combinations				
iHARsa	0.922	0.795	0.710	0.772
iHARwarq	0.894	0.810	0.736	0.745
sa	0.866	0.709	0.589	0.554
warq	0.835	0.691	0.580	0.543

TABLE B4 DJI forecasting performance for different approaches using the rolling window MSE ratios relative to the HAR model for the 2012–2018 period (low volatility).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000	1.000	1.000	1.000
HARQ	1.079	1.005	0.992	0.963
HARd	1.106	1.375	1.437	1.480
HARw	1.111	1.126	1.135	1.152
HARm	1.187	1.163	1.129	1.081
Forecasts based on lags				
RV^d_{t-1}	1.322	2.150	2.492	2.907
RV^w_{t-1}	1.155	1.277	1.398	1.544
RV^m_{t-1}	1.209	1.201	1.179	1.138
Forecast combinations				
iHARsa	1.029	1.077	1.101	1.135
iHARwarq	1.053	1.092	1.114	1.125
sa	0.981	1.081	1.144	1.268
warq	0.987	0.991	1.006	1.034

TABLE B5 DJI forecasting performance for different approaches using the rolling window MSE ratios relative to the HAR model for the 2019–2020 period (high volatility).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000	1.000	1.000	1.000
HARQ	1.072	0.993	1.142	1.080
HARd	1.028	0.542	0.600	0.999
HARw	1.047	1.125	1.210	1.050
HARm	2.001	1.949	1.827	2.102

(Continues)

TABLE B5 (Continued)

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts based on lags				
RV^d_{t-1}	1.056	0.592	0.567	1.793
RV^w_{t-1}	0.954	0.618	0.551	1.653
RV^m_{t-1}	1.926	1.103	0.815	1.797
Forecast combinations				
iHARsa	1.050	0.905	0.933	1.205
iHARwarq	1.033	0.838	0.815	1.119
sa	0.952	0.547	0.486	1.413
warq	0.995	0.552	0.469	1.264

TABLE B6 Forecasting results for the individual stocks. Forecasting performance is reported in the form of rolling window MSE ratios relative to the HAR model for the full sample, averaged across the sample of stocks. The number of times each forecast is included in the MCS at a 10% level of significance is reported in the superscripts.

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000^{25}	1.000^{26}	1.000^{26}	1.000^{26}
HARQ	1.591^{25}	1.061^{26}	1.208^{26}	1.459^{26}
HARd	1.169^{25}	1.089^{25}	0.991^{25}	1.011^{25}
HARw	0.957^{25}	1.009^{26}	0.846^{26}	0.905^{26}
HARm	1.089^{25}	1.228^{25}	1.329^{25}	1.508^{25}
Forecasts based on lags				
RV^d_{t-1}	1.208^{24}	1.504^{22}	1.550^{22}	1.762^{24}
RV^w_{t-1}	0.917^{25}	0.892^{25}	0.842^{25}	0.981^{26}
RV^m_{t-1}	1.056^{25}	0.962^{26}	0.835^{26}	0.879^{26}
Forecast combinations				
iHARsa	0.910^{25}	0.855^{26}	0.795^{26}	0.894^{26}
iHARwarq	0.895^{25}	0.850^{26}	0.810^{26}	0.884^{26}
sa	0.852^{25}	0.780^{26}	0.739^{26}	0.859^{26}
warq	0.823^{26}	0.718^{26}	0.678^{26}	0.772^{26}

TABLE B7 Forecasting results for the individual stocks. Forecasting performance is reported in the form of rolling window QLIKE ratios relative to the HAR model for the full sample, averaged across the sample of stocks. The number of times each forecast is included in the MCS at a 10% level of significance is reported in the superscripts.

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000^{16}	1.000^{12}	1.000^{13}	1.000^{20}
HARQ	1.091^{14}	0.958^{26}	0.970^{25}	1.026^{23}
HARd	1.265^{0}	1.410^{0}	1.407^{0}	1.378^{0}
HARw	1.124^{0}	1.129^{0}	1.100^{2}	1.108^{4}
HARm	1.318^{0}	1.244^{0}	1.175^{0}	1.121^{1}
Forecasts based on lags				
RV^d_{t-1}	1.255^{0}	1.575^{0}	1.709^{0}	1.823^{0}
RV^w_{t-1}	1.097^{0}	1.110^{0}	1.127^{0}	1.186^{0}
RV^m_{t-1}	1.303^{0}	1.213^{0}	1.138^{1}	1.075^{14}
Forecast combinations				
iHARsa	1.139^{0}	1.163^{0}	1.140^{0}	1.135^{1}
iHARwarq	1.145^{0}	1.188^{0}	1.175^{0}	1.175^{0}
sa	0.969^{26}	0.961^{25}	0.973^{20}	1.019^{22}
warq	1.003^{5}	1.063^{0}	1.113^{0}	1.191^{0}

TABLE B8 Forecasting results for the individual stocks. Forecasting performance is reported in the form of rolling window MSE ratios relative to the log-HAR model for the full sample, averaged across the sample of stocks. The number of times each forecast is included in the MCS at a 10% level of significance is reported in the superscripts.

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from log-HAR and related models				
log-HAR	1.000^{26}	1.000^{26}	1.000^{26}	1.000^{26}
log-HARd	1.189^{24}	1.392^{26}	1.396^{26}	1.292^{26}
log-HARw	1.084^{24}	1.106^{26}	1.091^{26}	1.067^{26}
log-HARm	1.327^{24}	1.333^{26}	1.239^{26}	1.135^{26}
Forecasts based on lags				
RV^d_{t-1}	1.448^{23}	1.879^{21}	2.030^{21}	2.045^{23}
RV^w_{t-1}	1.098^{25}	1.114^{26}	1.102^{26}	1.138^{26}
RV^m_{t-1}	1.263^{24}	1.204^{26}	1.102^{26}	1.029^{26}
Forecast combinations				
log-iHARsa	1.091^{25}	1.139^{26}	1.117^{26}	1.077^{26}
sa	1.020^{26}	0.974^{26}	0.969^{26}	0.999^{26}

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TABLE B9 Forecasting results for the individual stocks. Forecasting performance is reported in the form of rolling window QLIKE ratios relative to the HAR model for the full sample, averaged across the sample of stocks. The number of times each forecast is included in the MCS at a 10% level of significance is reported in the superscripts.

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
Forecasts from log-HAR and related models				
log-HAR	1.000^{26}	1.000^{26}	1.000^{26}	1.000^{26}
log-HARd	1.133^{0}	1.256^{0}	1.282^{0}	1.276^{0}
log-HARw	1.115^{0}	1.078^{3}	1.060^{14}	1.059^{15}
log-HARm	1.421^{0}	1.336^{0}	1.250^{0}	1.141^{2}
Forecasts based on lags				
RV^{d}_{t-1}	1.333^{0}	1.627^{0}	1.712^{0}	1.719^{0}
RV^{w}_{t-1}	1.166^{0}	1.146^{0}	1.131^{0}	1.118^{0}
RV^{m}_{t-1}	1.385^{0}	1.254^{0}	1.141^{4}	1.013^{24}
Forecast combinations				
log-iHARsa	1.069^{0}	1.052^{19}	1.046^{25}	1.049^{20}
sa	1.030^{5}	0.992^{26}	0.976^{26}	0.961^{25}

TABLE B10 Extra assets considered in the robustness check.

USO United States Oil ETF	10-Apr-2006 to 29-Dec-2022
GLD SPDR Gold Trust ETF	18-Nov-2004 to 29-Dec-2022
SLV, iShares Silver Trust ETF	28-Apr-2006 to 29-Dec-2022
EUR/USD	01-Feb-2000 to 01-Feb-2023
GBP/USD	01-Feb-2000 to 01-Feb-2023
AORD (Australia)	04-Jan-2000 to 28-Jun-2022
FTSE 100 (UK)	04-Jan-2000 to 28-Jun-2022
DAX (Germany)	03-Jan-2000 to 28-Jun-2022
HSI (Hong Kong)	03-Jan-2000 to 24-Jun-2022
N225 (Japan)	02-Feb-2000 to 28-Jun-2022
STOXX	03-Jan-2000 to 28-Jun-2022

TABLE B11 USO forecasting performance for different approaches using the rolling window MSE ratios relative to the HAR model for the whole 2000–2020 period. $*$ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^{d}_{t-1}, RV^{w}_{t-1}, and RV^{m}_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000*	1.000*	1.000*	1.000*
HARQ	1.049*	1.325*	1.090*	1.131*
HARd	1.023*	0.917*	1.232	1.228*
HARw	1.111*	0.977*	0.993*	1.082*
HARm	1.426*	1.164*	1.952*	1.780*
Forecasts based on lags				
RV^{d}_{t-1}	1.074*	1.109*	1.595	1.610*
RV^{w}_{t-1}	1.066*	0.755*	0.995*	1.129*
RV^{m}_{t-1}	1.398*	1.019*	1.420	1.387*
Forecast combinations				
iHARsa	0.960*	0.747*	1.056*	1.169*
iHARwarq	0.999*	0.759*	1.069*	1.146*
sa	0.906*	0.677*	0.946*	1.061*
warq	0.965*	0.683*	0.925*	1.001*

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TABLE B12 USO forecasting performance for different approaches using the rolling window QLIKE ratios relative to the HAR model for the whole 2000–2020 period. ∗ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000	1.000∗	1.000∗	1.000∗
HARQ	1.028	1.116	1.096	1.080∗
HARd	1.234	1.486	1.457	1.382
HARw	1.068	1.131	1.113	1.115
HARm	1.260	1.339	1.282	1.185
Forecasts based on lags				
RV^d_{t-1}	1.421	1.876	1.877	1.835
RV^w_{t-1}	1.065	1.102	1.106	1.148
RV^m_{t-1}	1.246	1.268	1.190	1.116∗
Forecast combinations				
iHARsa	1.050	1.128	1.124	1.124
iHARwarq	1.055	1.150	1.146	1.147
sa	0.974∗	0.971∗	0.974∗	1.020∗
warq	1.034	1.118	1.124	1.176

TABLE B13 SLV forecasting performance for different approaches using the rolling window MSE ratios relative to the HAR model for the whole 2000–2020 period. ∗ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000∗	1.000∗	1.000∗	1.000∗
HARQ	1.552	1.013∗	1.007∗	1.005∗

(Continues)

TABLE B13 (Continued)

	Forecasting horizon			
	1-day	5-day	10-day	22-day
HARd	1.021∗	1.080	1.098	1.144
HARw	1.067	1.067∗	1.069∗	1.123
HARm	1.250	1.215	1.164	1.055∗
Forecasts based on lags				
RV^d_{t-1}	1.217	1.853	2.230	2.866
RV^w_{t-1}	1.117	1.238	1.366	1.692
RV^m_{t-1}	1.265	1.237	1.212	1.207
Forecast combinations				
iHARsa	0.992∗	0.976∗	0.972∗	0.995∗
iHARwarq	1.015∗	0.993∗	0.979∗	0.990∗
sa	0.943∗	1.018∗	1.111	1.342
warq	0.961∗	0.973∗	1.018∗	1.161

TABLE B14 SLV forecasting performance for different approaches using the rolling window QLIKE ratios relative to the HAR model for the whole 2000–2020 period. ∗ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HARQ in Equation (5), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, iHARwarq—weighted average of the same models using weights based on RQ in Equation (8), sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7), warq—weighted average of the same lags in Equation (9).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000∗	1.000∗	1.000∗	1.000∗
HARQ	1.032∗	1.041∗	1.006∗	0.990∗
HARd	1.153	1.366	1.415	1.369
HARw	1.088	1.147	1.137	1.152
HARm	1.142	1.121	1.108∗	1.064∗
Forecasts based on lags				
RV^d_{t-1}	1.570	2.119	2.324	2.504
RV^w_{t-1}	1.136	1.228	1.234	1.380
RV^m_{t-1}	1.123	1.059∗	1.013∗	0.973∗
Forecast combinations				
iHARsa	1.055	1.114	1.122	1.119
iHARwarq	1.062∗	1.152∗	1.174	1.171
sa	1.013	1.036	1.028∗	1.116
warq	1.118	1.247	1.294	1.435

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TABLE B15 GBP forecasting performance for different approaches using the rolling window MSE ratios relative to the HAR model for the whole 2000–2020 period. $*$ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000$*$	1.000$*$	1.000$*$	1.000$*$
HARd	1.188$*$	0.762$*$	0.758$*$	0.548$*$
HARw	0.956$*$	1.041$*$	0.824$*$	0.584$*$
HARm	0.933$*$	0.703$*$	0.771$*$	1.028$*$
Forecasts based on lags				
RV^d_{t-1}	1.593$*$	2.229$*$	2.684$*$	2.794$*$
RV^w_{t-1}	1.023$*$	0.907$*$	0.940$*$	0.919$*$
RV^m_{t-1}	0.945$*$	0.676$*$	0.619$*$	0.528$*$
Forecast combinations				
iHARsa	0.960$*$	0.691$*$	0.619$*$	0.543$*$
sa	0.996$*$	0.856$*$	0.883$*$	0.850$*$

TABLE B16 GBP forecasting performance for different approaches using the rolling window QLIKE ratios relative to the HAR model for the whole 2000–2020 period. $*$ indicates that a forecast is included in the MCS at a 10% level of significance. Model specifications: HAR in Equation (4), HAR with only daily (HARd), weekly (HARw) or monthly (HARm) predictors in Equation (6). Forecast combinations: iHARsa—average of HAR models with individual predictors, sa—simple average of RV^d_{t-1}, RV^w_{t-1}, and RV^m_{t-1} in Equation (7).

	Forecasting horizon			
	1-day	5-day	10-day	22-day
Forecasts from HAR and related models				
HAR	1.000$*$	1.000$*$	1.000$*$	1.000$*$
HARd	1.282	1.560	1.557	1.508
HARw	1.053	1.097	1.077	1.084
HARm	1.049	1.066	1.060	1.048
Forecasts based on lags				
RV^d_{t-1}	17.131	20.236	19.702	17.028
RV^w_{t-1}	1.025$*$	1.057$*$	1.081	1.200
RV^m_{t-1}	1.035$*$	1.043$*$	1.056$*$	1.058$*$
Forecast combinations				
iHARsa	1.066	1.128	1.124	1.126
sa	0.989$*$	0.998$*$	1.017$*$	1.115